

WEEK 6



Redstone Levers

Flip a lever, open a door

레버를 당겨 레드스톤 문 열기 (영어코딩)

BIG PICTURE

The agent can press things



A lever sends a redstone signal that opens a closed door.

레버 신호가 닫힌 문을 열어줘요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is `interact` ?

`agent.interact` presses or flips whatever the agent faces.

```
agent.interact(FORWARD)
# flips the lever right in front
```

Be standing **right in front** of the lever first.

앞에 있는 레버를 작동시켜요

CONCEPT

Order is everything



flip first

door opens
then walk through



walk first

door is shut
agent gets stuck

먼저 레버, 그다음 통과

CODE

Flip then pass through

```
agent.move(FORWARD, 1)
agent.interact(FORWARD)    # Lever ON, door opens
agent.move(FORWARD, 2)    # walk through
```

Reach the lever, flip it, then go through the now-open door.

레버까지 가서 당기고 문을 통과해요

WATCH IT!

Step by step

`agent.move(FORWARD, 1)`



`agent.interact(FORWARD)`



lever ON → redstone signal



 door opens ✓

CODE

Walk to the lever first

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)  
agent.interact(FORWARD)  
agent.turn(RIGHT_TURN)  
agent.move(FORWARD, 2)
```

Loop up to the wall, flip the lever, turn, and continue.

PREDICT 🤖

What happens to the door?

Predict first — then click reveal

```
agent.move(FORWARD, 1)
agent.interact(FORWARD)
agent.move(FORWARD, 1)
```

Reveal output 🔍 reveal

VOCAB

New words today

lever a switch that sends a redstone signal

interact press or flip what the agent faces

redstone Minecraft's wiring that carries signals

signal the power that opens the door

레버

작동시키다

레드스톤

신호



DEBUG

What's wrong with this code?

Goal: flip the lever, then pass the door.

buggy

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(FORWARD)
```

Think first — then click next.



DEBUG

The bug

Bug: the agent walks into the closed door *before* flipping the lever, so it gets stuck. Flip the lever at the first wall, then move on.

fixed

```
agent.move(FORWARD, 1)
agent.interact(FORWARD)
agent.move(FORWARD, 1)
```

YOUR TURN!

Open the door

- ✓ Open the Week 6 world + Code Builder
- ✓ Move the agent up to the lever
- ✓ Flip it with `agent.interact(FORWARD)`
- ✓ Walk through the open door to the goal
- ✓ Send your walkthrough video on KakaoTalk



QUIZ

QUESTION 1 OF 3

What does `agent.interact(FORWARD)` do?

Presses or flips the block right in front of the agent.

Places a stone block.

Moves the agent two blocks.

Ends the program.



QUIZ

QUESTION 2 OF 3

Before flipping a lever, the agent must be...

Far away from it.

Facing the ceiling.

Holding a block.

Standing right in front of it.



QUIZ

QUESTION 3 OF 3

Why flip the lever **before** moving forward?

Levers only work while moving.

The door is closed until the lever is flipped — walk first and you get stuck.

It looks neater.

It doesn't matter.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Lever flipped!

Next: power pistons to cross gaps.

다음 주: 피스톤으로 다리 만들기